EXPERIMENT 2

Objective 1:

```
clc;

 clear;

 close all;

% Time Vector t = -1:0.001:1;

% Input Signal x = sin(2pi5t) + 0.5cos(2pi3*t);

% Time-Reversed Signal xr = fliplr(x);

% Even and Odd Components xe = (x + xr)/2; xo = (x - xr)/2;

% Reconstructed Signal xrec = xe + xo;

% Plotting figure;

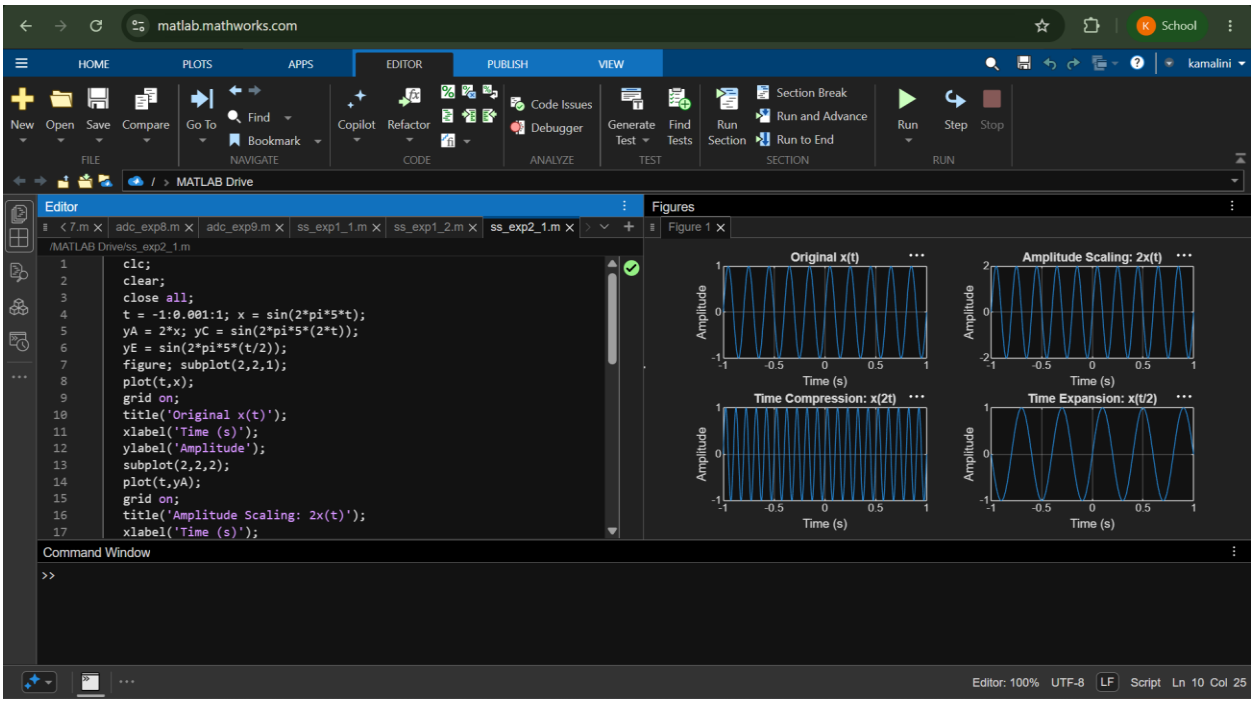
subplot(2,2,1); plot(t, x); grid on; title('Input x(t)'); xlabel('Time (s)'); ylabel('Amplitude');

subplot(2,2,2); plot(t, xe); grid on; title('Even Component'); xlabel('Time (s)');
ylabel('Amplitude');

subplot(2,2,3); plot(t, xo); grid on; title('Odd Component'); xlabel('Time (s)');
ylabel('Amplitude');

subplot(2,2,4); plot(t, xrec); grid on; title('Reconstructed x(t)'); xlabel('Time (s)');
ylabel('Amplitude');

% Display Reconstruction Error fprintf('Maximum reconstruction error = %g\n', max(abs(x -
xrec)));
```



Objecvtive 2:

clc; clear; close all;

% Time Vector t = -1:0.001:1;

% Original Signal x = sin(2*pi*5*t);

% Amplitude Scaling yA = 2*x;

% Time Compression yC = sin(2*pi*5*(2*t));

% Time Expansion yE = sin(2*pi*5*(t/2));

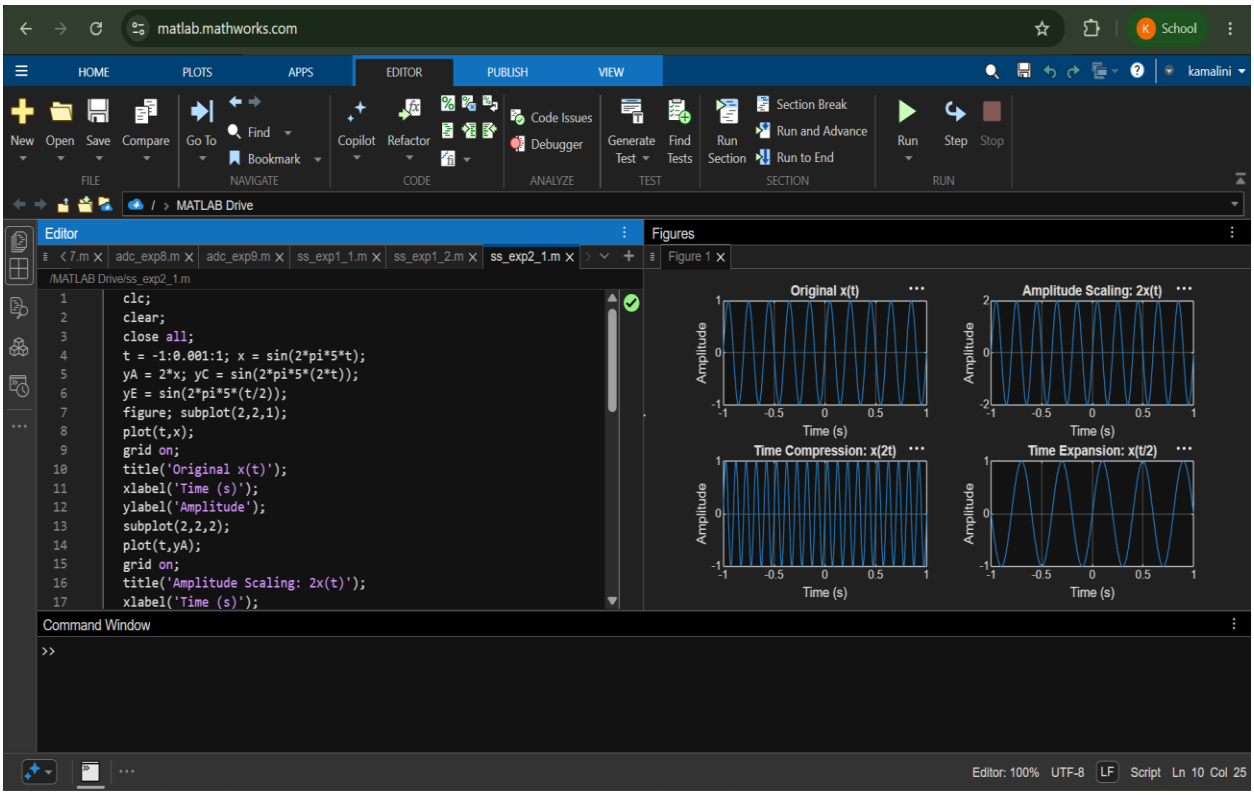
% Plotting Signals figure;

subplot(2,2,1); plot(t, x); grid on; title('Original x(t)'); xlabel('Time (s)'); ylabel('Amplitude');

subplot(2,2,2); plot(t, yA); grid on; title('Amplitude Scaling: 2x(t)'); xlabel('Time (s)'); ylabel('Amplitude');

subplot(2,2,3); plot(t, yC); grid on; title('Time Compression: x(2t)'); xlabel('Time (s)'); ylabel('Amplitude');

subplot(2,2,4); plot(t, yE); grid on; title('Time Expansion: x(t/2)'); xlabel('Time (s)'); ylabel('Amplitude');


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SECTION
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adc_exp9.m
ss_exp1_1.m
ss_exp1_2.m
ss_exp2_1.m
/MATLAB Drive/ss_exp2_1.m
clc;
clear;
close all;
t = -1:0.001:1; x = sin(2*pi*5*t);
yA = 2*x; yC = sin(2*pi*5*(2*t));
yE = sin(2*pi*5*(t/2));
figure; subplot(2,2,1);
plot(t,x);
grid on;
title('Original x(t)');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(2,2,2);
plot(t,yA);
grid on;
title('Amplitude Scaling: 2x(t)');
xlabel('Time (s)');
Figures
Figure 1
Original x(t)
Amplitude Scaling: 2x(t)
Time Compression: x(2t)
Time Expansion: x(t/2)
Amplitude
Time (s)
Command Window
>>
Editor: 100%
UTF-8
LF
Script
Ln 10 Col 25